

ZKTrustLLM-Agents L4: n8n-Orchestrated Scientific Evaluation of Proof-Governed Agentic Zero-Trust Control for O-RAN Edge Security

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Abstract—Autonomous O-RAN security operations require policy-governed action, auditable evidence, replay-resistant anchoring, and reproducible evaluation. This paper presents ZKTrustLLM-Agents L4, a proof-governed agentic zero-trust framework for accountable O-RAN edge security workflows. The framework decomposes autonomy into bounded telemetry, reasoning, policy, proof, and audit/anchor agents. A self-hosted n8n workflow orchestrates scientific evaluation and generates run identifiers, Git commit binding, evidence hashes, raw logs, and paper-ready KPI tables. A clean 240-record dataset is produced across six variants, four impairment profiles, and ten repetitions per condition. Stage 3 direct runtime hooks populate anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection. A consensus/hash extension separates measured Sepolia-IoTeX first inclusion from sourced Raft/QBFT protocol properties and adds a gated comparison of complete SHA-256 and SHA-512 digest anchoring. The result is a reviewer-verifiable evaluation method for accountable autonomous O-RAN security rather than an unconstrained agentic dashboard.

Index Terms—O-RAN, zero trust, agentic AI, n8n, zero-knowledge proof, blockchain audit, consensus, SHA-2, reproducible evaluation.

I. INTRODUCTION

O-RAN disaggregates the radio access network into programmable components, including the SMO, Non-RT RIC, Near-RT RIC, rApps, and xApps. This programmability enables closed-loop optimisation but also creates security risks: unsafe autonomous action, weak evidence binding, replayed audit records, and insufficient reproducibility of security experiments.

This paper asks: *how can autonomous O-RAN security actions be made policy-gated, evidence-bound, replay-resistant, and scientifically reproducible?* We answer this through ZKTrustLLM-Agents L4, a proof-governed agentic zero-trust framework.

The novelty is not the isolated use of n8n, blockchain, IPFS, or agentic AI. The novelty is their controlled scientific composition: bounded agents reason over telemetry, policy gates restrict actions, proof paths attest admissibility, audit hooks produce direct runtime evidence, and n8n orchestrates repeatable scientific evaluation. The design is motivated by zero-trust principles [1], O-RAN policy-control workflows [2], succinct proof systems such as Groth16 [3], self-hosted

TABLE I
BOUNDED AGENTIC COMPONENTS

Agent	Role	Output
Telemetry	Observe O-RAN/media KPIs	Jitter, loss, profile
Reasoning	Map state to trust/action	Trust state, action class
Policy	Enforce action ladder	Auto/Human/Privileged/Never
Proof	Check admissibility	AUTH_V2.x status
Audit/Anchor	Bind evidence and commit	CID/hash/anchor record

command orchestration through n8n [4], and the public implementation branch used for this evaluation [5].

A. Contributions

The paper makes seven contributions: (i) a five-agent O-RAN security architecture; (ii) a policy-gated OBSERVE–REASON–PROVE–ANCHOR–ACT loop; (iii) an n8n-orchestrated evaluation pipeline; (iv) a clean 240-record dataset; (v) Stage 3 direct runtime hooks for anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection; (vi) a fail-closed, paired public-testnet protocol for comparing identical audit-anchor bytecode on Ethereum Sepolia (Gasper PoS) and IoTeX testnet (Roll-DPoS/PBFT), without introducing unmeasured values; and (vii) a supervisor-requested consensus/hash extension that distinguishes measured public-testnet evidence from Raft/QBFT protocol properties and validates complete SHA-256/SHA-512 digest anchoring.

II. ARCHITECTURE AND THREAT MODEL

Fig. 1 shows the proposed architecture. The system places MNO policy at the SMO/Non-RT RIC layer and uses bounded agents for O-RAN security automation. The telemetry agent observes network and media-plane indicators. The reasoning agent maps state to trust level and candidate action. The policy agent enforces the action ladder. The proof agent supports bounded admissibility checks. The audit/anchor agent writes evidence hashes and compact commitments.

A. Threat Controls

The framework considers unapproved autonomous action, evidence tampering, anchor replay, zero/empty anchors, unsafe escalation, and unauthorized submitters. Fig. 2 maps these concepts to controls.

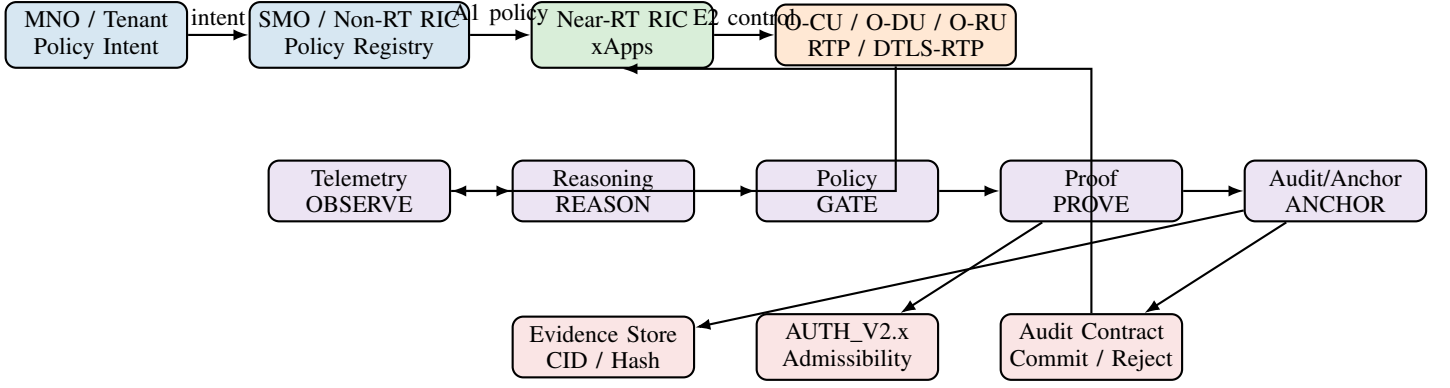


Fig. 1. Proposed ZKTrustLLM-Agents L4 architecture. Policy is placed in the SMO/Non-RT RIC governance path while bounded agents connect O-RAN telemetry to proof, evidence, and audit anchoring.

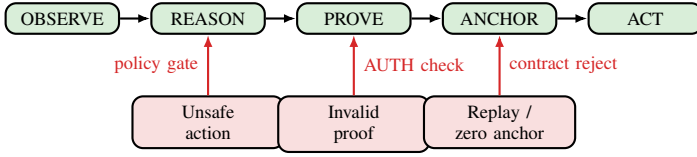


Fig. 2. Threat-to-control mapping. Unsafe actions are policy-gated, invalid proofs are blocked, and replay/zero anchors are rejected.

B. Smart-Contract Boundary

The smart contract does not control O-RAN traffic directly. Its role is trust-plane accountability: storing compact commitments, rejecting duplicate commitments, rejecting zero commitments, and enforcing authorized submitters. It cannot verify physical network truth, guarantee evidence availability, or replace MNO operational policy.

III. SCIENTIFIC EVALUATION METHODOLOGY

The evaluation is organised as a three-stage pipeline. Stage 1 establishes the n8n workflow, response matrix, KPI plan, and formal-verification preparation. Stage 2 converts outputs into a clean 240-record dataset. Stage 3 adds direct runtime hooks.

A. Evaluation Matrix

The experiment covers six variants: Full L4, No-ZK, Oracle-only, RBAC-only, No-IPFS, and No-policy-gate. Each variant is evaluated over four impairment profiles and ten repetitions, giving 240 clean records.

B. KPI Classes

The dataset distinguishes direct runtime KPIs, configured impairment-profile KPIs, and missing values. This prevents overclaiming. Configured RTP/DTLS-RTP jitter and loss values are scenario-control parameters until replaced by packet-capture or O-RAN telemetry measurements.

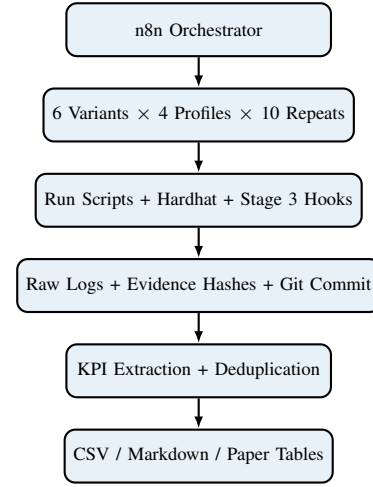


Fig. 3. n8n-orchestrated scientific evaluation pipeline. n8n coordinates execution but is not a security primitive.

C. Consensus-Layer Sensitivity Protocol

The consensus mechanism belongs to the execution network, not to the Solidity contract. We therefore add a paired public-testnet protocol that deploys identical Stage3AnomalyLedger bytecode to Ethereum Sepolia and IoTeX testnet. Sepolia follows the Ethereum PoS protocol but uses a permissioned validator set controlled by client and testing teams [6]; IoTeX documents Randomized Delegated Proof of Stake (Roll-DPoS), with delegated candidate election, VRF committee selection, and PBFT voting [7]. IoTeX testnet is EVM compatible and uses chain ID 4690 [8].

Each session executes three excluded warm-up pairs and 30 measured pairs. A matched pair carries identical commitment and evidence digests, with both transactions submitted concurrently from one client host. The publication gate requires at least three sessions, verifies chain identities and runtime-code hashes, and rejects incomplete receipts, events, block samples, or security probes. The primary metric is client-observed submission-to-first-inclusion latency; it includes RPC/client overhead and public-testnet load and is not economic final-

TABLE II
NOVELTY AND COMPARISON MATRIX

Approach	Agents	Policy	ZK	Evidence	Audit	Replay	KPIs
O-RAN monitoring only	No	Partial	No	No	No	No	Network only
Blockchain logging only	No	No	No	Partial	Yes	Partial	Gas/tx
RBAC contract only	No	Yes	No	No	Yes	Partial	Auth/gas
Oracle-only trust	Yes	Weak	No	Partial	Yes	No	Score/gas
No-ZK ablation	Yes	Yes	No	Yes	Yes	Yes	Workflow/gas
Full L4	Yes	Yes	Yes	Yes	Yes	Yes	Agent/ZK/audit
n8n-orchestrated L4	Yes	Yes	Optional	Yes	Yes	Yes	240 records

ity. Median confidence intervals use a two-level hierarchical bootstrap over sessions and within-session pairs. Gas prices are recorded but not converted into a cross-asset cost comparison.

D. Consensus-Protocol and SHA-2 Extension

Consensus and evidence hashing are separate layers. Ethereum uses the Gasper consensus mechanism, whereas IoTeX uses Roll-DPoS with PBFT-style finalisation [7], [9]. Figure 4 therefore keeps the existing Sepolia and IoTeX measurements distinct from Raft, a crash-fault-tolerant leader/majority protocol [10], [11], and QBFT, a Byzantine-fault-tolerant permissioned EVM protocol requiring at least a two-thirds validator supermajority [12]. The latter two rows are protocol-property comparisons unless raw deployment evidence passes the same publication gate. The measured latency figure remains restricted to the 90 matched Sepolia–IoTeX pairs; Raft and QBFT are not inserted as synthetic bars.

SHA-256 and SHA-512 are one-way SHA-2 hash functions, not encryption algorithms [13]. Table IV and Fig. 5 define a fair comparison: identical retained payload bytes are hashed off chain and the complete 32-byte or 64-byte digest is anchored through identical `HashCommitmentLedger` bytecode. This measures digest anchoring. Solidity exposes SHA-256 through a precompile-backed built-in, but the standard EVM has no SHA-512 precompile [14]; the separate on-chain SHA-256 path is therefore not misrepresented as a symmetric SHA-256/SHA-512 computation benchmark.

The separate local primitive microbenchmark uses one isolated Hardhat contract to compare the Keccak-256 opcode path, the SHA-256 precompile path, and a FIPS-vector-gated pure-Solidity SHA-512 reference. It evaluates deterministic 32-byte and 1,024-byte inputs, with three excluded warm-ups and 30 retained `eth_call` observations per algorithm/input cell. Algorithm order is cyclically balanced. The measured quantity is the `gasleft()` delta around the primitive path under Solidity 0.8.20, optimizer 200, `viaIR=true`, and the Paris EVM target. This local instrument is not the deployable SHA-512 path and cannot support a consensus, latency, finality, fee, or public-testnet claim.

IV. RESULTS

A. KPI Completeness

Table V summarises KPI completeness after Stage 3. Anchor gas, reasoning latency, replay rejection, zero-anchor rejection, and unauthorized-submitter rejection are populated from direct runtime hooks. RTP/DTLS-RTP jitter and loss are configured impairment-profile values.

B. Variant/Profile Summary

Table VI summarises the clean 240-record evaluation. The RBAC-only baseline was repaired to match the current contract ABI and rerun only for the affected variant, preserving experimental traceability.

C. Stage 3 Runtime Evidence

Stage 3 introduces an isolated `Stage3AnomalyLedger` micro-benchmark and a deterministic policy-reasoning probe. These emit parseable KPI lines for anchor gas, negative-security rejection, and reasoning latency.

D. PoS–Roll-DPoS Consensus Sensitivity

Across 90 matched transaction pairs collected in 3 independent sessions, median client-observed first-inclusion latency was 12426.58 ms on Ethereum Sepolia and 7225.15 ms on IoTeX testnet (Table VIII). The median paired Sepolia-to-IoTeX latency ratio was 1.87 (hierarchical-bootstrap 95% CI: 1.48–2.74). The median anchor execution gas was 48973 and 43685, respectively; gas-price values are not converted into a cross-asset economic comparison. All three read-only rejection probes (duplicate, zero, and unauthorised submitter) reverted on both deployed contracts. These values describe public-testnet first inclusion for the isolated audit-anchor transaction and do not imply mainnet finality or production O-RAN performance.

E. Local Hash-Primitive Microbenchmark

The local hash publication gate passed all retained-payload digest checks and the SHA-256/SHA-512 validation vectors. For the 32-byte input, the median primitive-path deltas were 59 gas for Keccak-256, 392 gas for the SHA-256 precompile path, and 162,174 gas for the pure-Solidity SHA-512 reference. For the 1,024-byte input, the median primitive-path deltas

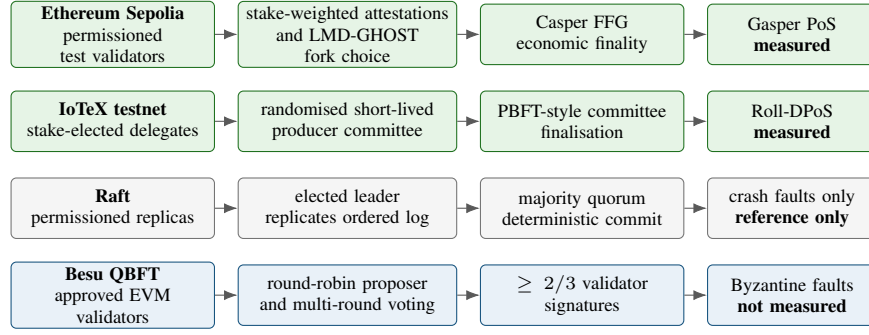


Fig. 4. Consensus-protocol comparison requested by the supervisor. Green lanes are the two public testnets with measured first-inclusion evidence. Raft is a crash-fault-tolerant replicated-log reference and is not an EVM public-testnet measurement. QBFT is the selected Byzantine-fault-tolerant EVM extension target, but no latency value is reported until an actual multi-validator Besu run passes the same evidence gate.

TABLE III
PROTOCOL-FAMILY COMPARISON SEPARATED FROM THE MEASURED LATENCY CHART. ONLY SEPOLIA AND IOTEX HAVE MATCHED DEPLOYMENT EVIDENCE IN THIS WORK [7], [9], [10], [12].

Protocol/deployment	Family	Fault/adversary model	Ordering or commit pattern	Evidence status in this work
Ethereum Sepolia	Gasper PoS	Stake-weighted Byzantine/economic assumptions	LMD-GHOST fork choice plus Casper FFG finality	90 matched public-testnet pairs
IoTeX testnet	Roll-DPoS	Byzantine delegate committee	Stake election, random committee, PBFT-style voting	90 matched public-testnet pairs
Raft	Leader/majority CFT	Crash faults; not Byzantine faults	Elected leader replicates a log to a majority	Protocol reference only
Besu QBFT	Permissioned BFT	Fewer than one-third Byzantine validators	Proposer plus prepare/commit supermajority voting	Extension target; not measured

TABLE IV
SHA-256 AND SHA-512 COMPARISON USED BY THE EVIDENCE-ANCHOR EXTENSION. BOTH ARE ONE-WAY SHA-2 HASH FUNCTIONS, NOT ENCRYPTION ALGORITHMS [13].

Algorithm	Digest (bits)	Message block (bits)	Word size (bits)	Generic collision work (bits, approx.)	EVM execution path	Benchmark interpretation
SHA-256	256	512	32	128	Standard SHA-256 precompile; off-chain mode also supported	Anchor one 32-byte full digest
SHA-512	512	1024	64	256	No standard SHA-512 precompile; compute off chain	Anchor one 64-byte full digest

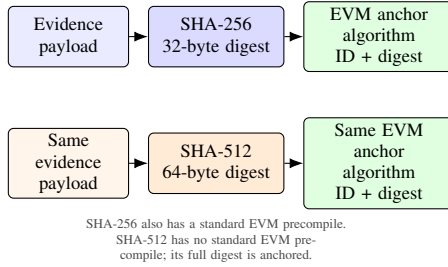


Fig. 5. Fair SHA-2 comparison boundary. Both algorithms hash identical bytes off chain and anchor their complete digest on identical EVM bytecode. The resulting network comparison measures digest anchoring, not encryption or CPU hash throughput.

TABLE V
KPI COMPLETENESS AFTER STAGE 3

KPI	Coverage	Evidence Type
prover_time_ms	160/240	ablation/log gap
anchor_gas	240/240	direct runtime
rtp_jitter_ms	240/240	configured profile
rtp_loss_pct	240/240	configured profile
reason_latency_ms	240/240	direct runtime
replay_rejected	240/240	direct runtime
zero_anchor_rejected	240/240	direct runtime
unauthorized_submitter	240/240	direct runtime
postAutoScore_gas	240/240	Hardhat log

were 245 gas for Keccak-256, 2,879 gas for the SHA-256 precompile path, and 1,614,065 gas for the pure-Solidity SHA-512 reference. All 30 calls per algorithm/input cell returned the same delta because the bytecode, input, and local EVM state were fixed. These values characterise this isolated execution path; they are not public-testnet latency, monetary cost, finality, or consensus measurements.

V. DISCUSSION AND LIMITATIONS

The framework intentionally avoids overclaiming. Stage3AnomalyLedger is an isolated scientific micro-

benchmark, not the production trust ledger. The deterministic reasoning probe measures bounded control-plane reasoning latency, not open-ended semantic LLM reasoning. Configured impairment-profile values are not packet-capture measurements.

The main remaining direct-runtime KPI is full ZK prover timing for full-ZK rows. Current prover-time values are populated for ablation rows where the prover is intentionally not invoked, while direct prover logs are still required for full AUTH_V2.x rows.

TABLE VI
CLEAN 240-RECORD VARIANT/PROFILE SUMMARY

Variant	Profile	Runs	Pass	Fail	Mean Duration (ms)
full_l4	clean_baseline	10	10	0	39055.4
full_l4	delay_20ms	10	10	0	47435.9
full_l4	delay_20ms_jitter_5ms	10	10	0	56881.0
full_l4	delay_30ms_jitter_10ms_loss_1pct	10	10	0	66822.6
no_ipfs	clean_baseline	10	10	0	1782.7
no_ipfs	delay_20ms	10	10	0	1742.6
no_ipfs	delay_20ms_jitter_5ms	10	10	0	1759.8
no_ipfs	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1736.5
no_policy_gate	clean_baseline	10	10	0	1741.3
no_policy_gate	delay_20ms	10	10	0	1769.1
no_policy_gate	delay_20ms_jitter_5ms	10	10	0	1739.7
no_policy_gate	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1743.2
no_zk	clean_baseline	10	10	0	1765.1
no_zk	delay_20ms	10	10	0	1754.2
no_zk	delay_20ms_jitter_5ms	10	10	0	1727.6
no_zk	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1789.3
oracle_only	clean_baseline	10	10	0	1777.8
oracle_only	delay_20ms	10	10	0	1752.8
oracle_only	delay_20ms_jitter_5ms	10	10	0	1783.4
oracle_only	delay_30ms_jitter_10ms_loss_1pct	10	10	0	1752.5
rbac_only	clean_baseline	10	10	0	3311.4
rbac_only	delay_20ms	10	10	0	3579.0
rbac_only	delay_20ms_jitter_5ms	10	10	0	3606.8
rbac_only	delay_30ms_jitter_10ms_loss_1pct	10	10	0	3606.6

TABLE VII
STAGE 3 DIRECT RUNTIME HOOK KPIS

KPI	Coverage	Result
Anchor gas	240/240	48961 gas
Reason latency	240/240	4.889 ms
Replay rejection	240/240	true
Zero-anchor rejection	240/240	true
Unauthorized submitter	240/240	true

The PoS–Roll-DPoS extension is a consensus-sensitivity micro-benchmark rather than a claim of consensus superiority. Sepolia’s validator set is permissioned, both networks are testnets, and first-inclusion latency also reflects RPC implementation, client location, propagation, and transient load. It cannot be interpreted as mainnet throughput, decentralisation, or economic-finality performance. Causal attribution beyond the directly observed latency, gas, receipt, event, rejection, and block-timestamp evidence is therefore excluded.

The added Raft and QBFT lanes are not appended to Fig. 6

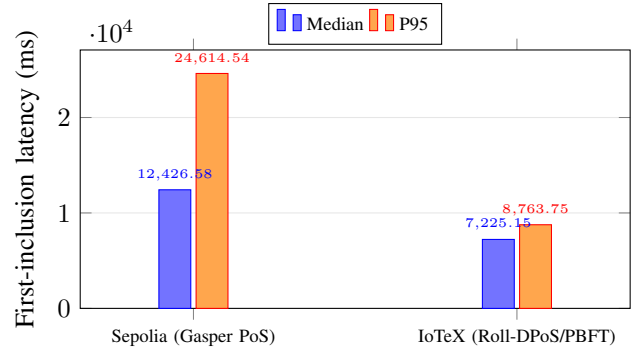


Fig. 6. Measured client-observed first-inclusion latency for matched, concurrently submitted audit-anchor transactions. Public-testnet load and RPC/client overhead are included; economic finality is not measured. Raft and QBFT are compared structurally in Fig. 4 but are omitted here because no corresponding measured deployment evidence exists.

as synthetic bars. Raft has a crash-fault model and a non-EVM Fabric deployment path in the cited reference, while QBFT requires a genuine multi-validator permissioned deployment. Both remain sourced protocol-property comparisons until mea-

TABLE VIII

PAIRED PUBLIC-TESTNET CONSENSUS SENSITIVITY OF THE IDENTICAL STAGE3ANOMALYLEDGER BYTECODE. LATENCY IS CLIENT-OBSERVED SUBMISSION TO FIRST INCLUSION, NOT ECONOMIC FINALITY; BRACKETS ARE HIERARCHICAL-BOOTSTRAP 95% CONFIDENCE INTERVALS FOR THE MEDIAN.

Network (consensus)	Chain ID	Anchor gas	Median latency (ms)	P95 latency (ms)	Block interval (s)	Rejections
Ethereum Sepolia (Gasper PoS)	11155111	48973	12426.58 [12394.75, 12455.95]	24614.54	12.00	3/3
IoTeX testnet (Roll-DPoS)	4690	43685	7225.15 [4538.58, 8493.93]	8763.75	3.00	3/3

TABLE IX
VALIDATED LOCAL HASH-PRIMITIVE MEASUREMENTS.

Primitive	Bytes	Median gas	n	Fixed
Keccak-256	32	59	30	yes
SHA-256	32	392	30	yes
SHA-512 (Solidity)	32	162,174	30	yes
Keccak-256	1,024	245	30	yes
SHA-256	1,024	2,879	30	yes
SHA-512 (Solidity)	1,024	1,614,065	30	yes

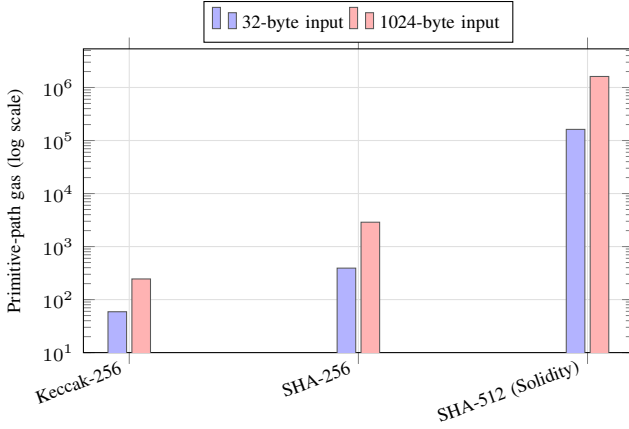


Fig. 7. Local Hardhat EVM hash-primitive microbenchmark. Values are `gasleft()` deltas around the isolated hash path under Solidity 0.8.20, optimizer 200, `viaIR=true`, and the Paris EVM target. The SHA-512 bar is a FIPS-vector-gated pure-Solidity reference, not the deployable evidence path. It is not a public-testnet latency, fee, finality, or consensus result.

sured under an explicit environment. Similarly, SHA-256 and SHA-512 results compare anchoring of complete precomputed digests. They do not imply that SHA is encryption, that SHA-512 was executed by a standard EVM precompile, or that a wider digest caused a consensus-level latency difference.

The local hash-primitive figure is also isolated from Fig. 6. Its `gasleft()` deltas characterise fixed bytecode on one Hardhat EVM. They do not establish cross-chain gas equality because execution-fork and network configurations must also match. The pure-Solidity SHA-512 reference is retained only to expose the absence of a standard SHA-512 precompile; the deployable workflow computes SHA-512 off chain and anchors the complete 64-byte digest.

VI. CONCLUSION

This paper presented ZKTrustLLM-Agents L4, a proof-governed and policy-gated agentic framework for accountable

O-RAN edge security workflows. The contribution is a reproducible scientific evaluation pipeline that turns an autonomous dashboard into reviewer-verifiable evidence. The clean 240-record dataset, Stage 3 runtime hooks, architecture, KPI completeness report, four-family consensus comparison, local hash-primitive reference, and evidence-gated SHA-2 digest-anchoring extension provide a foundation for conference and journal-level validation without converting unmeasured protocol profiles or local execution paths into public-testnet claims.

REFERENCES

- [1] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, “Zero trust architecture,” National Institute of Standards and Technology, Tech. Rep. NIST SP 800-207, 2020.
- [2] “O-ran sc al mediator documentation,” <https://docs.o-ran-sc.org/projects/o-ran-sc-ric-plt-a1/en/latest/overview.html>, 2026.
- [3] J. Groth, “On the size of pairing-based non-interactive arguments,” in *EUROCRYPT*, 2016, pp. 305–326.
- [4] “n8n execute command node documentation,” <https://docs.n8n.io/integrations/builtin/core-nodes/n8n-nodes-base-executecommand/>, 2026.
- [5] S. Karmaker, “Zktrustllm-agents l4 repository and supervisor 258 evaluation branch,” <https://github.com/Sudarson2021/zktrustllm-agents-l4/pull/1>, 2026.
- [6] Ethereum Foundation, “Ethereum networks: Sepolia,” <https://ethereum.org/developers/docs/networks/>, accessed: 2026-07-21.
- [7] IoTeX Foundation, “Randomized delegated proof of stake (roll-dpos) consensus,” <https://docs.iotex.io/blockchain/learn-iotex/core-concepts/consensus-mechanism>, accessed: 2026-07-21.
- [8] —, “Iotex testnet and evm integration parameters,” <https://docs.iotex.io/blockchain/learn-iotex/exchange-integration>, accessed: 2026-07-21.
- [9] Ethereum Foundation, “Consensus mechanisms: Gasper,” <https://ethereum.org/developers/docs/consensus-mechanisms/>, accessed: 2026-07-25.
- [10] D. Ongaro and J. Ousterhout, “In search of an understandable consensus algorithm,” in *2014 USENIX Annual Technical Conference*, 2014, pp. 305–319.
- [11] Hyperledger Foundation, “Hyperledger fabric ordering service: Raft,” https://hyperledger-fabric.readthedocs.io/en/latest/orderer/ordering_service.html, accessed: 2026-07-25.
- [12] Hyperledger Besu, “Configure qbft consensus,” <https://besu.hyperledger.org/private-networks/how-to/configure/consensus/qbft>, accessed: 2026-07-25.
- [13] National Institute of Standards and Technology, “Secure hash standard,” NIST, Tech. Rep. FIPS PUB 180-4, 2015.
- [14] Solidity Team, “Units and globally available variables: Cryptographic functions,” <https://docs.soliditylang.org/en/latest/units-and-global-variables.html#mathematical-and-cryptographic-functions>, accessed: 2026-07-25.